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Model-X Knockoffs
Panning for Gold (2018)
Metropolized (2021)
New perspectives (2023)

Linear Algebra
Methodology
Model Tests
Real Data

Model-X Knockoffs

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Presentation Overview

1. About Me
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   - Panning for Gold (2018)
   - Metropolized (2021)
   - New perspectives (2023)
3. Linear Algebra
   - Methodology
   - Model Tests
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About Me

- Studied Mathematics with German at Imperial College, London.
- In 2017, completed a PhD in Finance at the University of Geneva as part of the Geneva Finance Research Institute where I specialised in statistical modelling, econometrics, behavioural economics, and asset pricing.
- Won the Swiss Finance Institute prize for Best Doctoral Paper in 2016.

About Me

- Became interested in data science and joined a tech startup in Geneva in 2018 where I learned to program in Python.
- Began freelancing as a data science consultant as part of the Toptal global platform in 2021 with whom I've worked on a range of data science and data engineering projects.
- For the last year, I've been doing research for HEIG-VD in Yverdon-les-Bains as part of a job placement scheme.
- Earned my Professional Data Engineer certification from Google Cloud in September 2023.

- Panning for Gold: Model-X Knockoffs for High-dimensional Controlled Variable Selection
  - Candès, Fan, Janson, Lv
  - Journal of the Royal Statistical Society Series B: Statistical Methodology 80, no. 3 (2018)
- Metropolized Knockoff Sampling
  - Bates, Candès, Janson, Wang
  - Journal of the American Statistical Association 116:535 (2021)
- New perspectives on knockoffs construction
  - Berti, Dreassi, Leisen, Pratelli
  - Journal of Statistical Planning and Inference 223 (2023)

Panning for Gold

How can one control for the false discovery rate in high-dimensional non-linear models?

What happens if you have more features than observations and yet you know that only a handful of them contain any explanatory power? For example, which genes affect a particular health outcome?

Model-X Knockoffs

There are n **independently and identically distributed** observations of (X, Y) where

$$X = (X_1, \ldots, X_p) \in \mathbb{R}^p \text{ and } Y \in \mathbb{R}$$

A new family of random variables, $\tilde{X} = \left(\tilde{X}_1, \ldots, \tilde{X}_p\right)$ is a knockoff of X if

1. for any subset $S \subset \{1, \ldots, p\}$,

$$\left(X, \tilde{X}\right)_{\text{swap}(S)} \stackrel{d}{=} \left(X, \tilde{X}\right)$$

2. $\tilde{X} \perp\!\!\!\perp Y | X$ if there is a response Y.

Example

If $p = 3$ and $S = \{2, 3\}$, then

$$\left(X_1, X_2, X_3, \tilde{X}_1, \tilde{X}_2, \tilde{X}_3\right)_{\text{swap}(\{2,3\})} \stackrel{d}{=} \left(X_1, \tilde{X}_2, \tilde{X}_3, \tilde{X}_1, X_2, X_3\right)$$

Normal $(0, \Sigma)$

Suppose

$$X \sim \mathcal{N}(0, \Sigma)$$

Then

$$\left(X, \tilde{X}\right) \sim \mathcal{N}(0, \boldsymbol{G}), \text{ where } \boldsymbol{G} = \begin{bmatrix} \Sigma & \Sigma - D \\ \Sigma - D & \Sigma \end{bmatrix}$$

where D is any diagonal matrix such that $\boldsymbol{G}$ is positive semidefinite.

Sampling Knockoffs

It is now possible to sample the knockoffs, $\tilde{X}$, from the conditional distribution

$$\tilde{X}|X \stackrel{d}{=} \mathcal{N}(\mu, \boldsymbol{V})$$

where

$$\begin{aligned} \mu &= X - X\Sigma^{-1}D \\ \boldsymbol{V} &= 2D - D\Sigma^{-1}D \end{aligned}$$

What if Non-Gaussian?

$j = 1$ **while** $j \leq p$ **do**
| Sample $\tilde{X}_j$ from $\mathcal{L}\left(X_j \middle| X_{-j}, \tilde{X}_{1:j-1}\right)$
| $j = j + 1$
end

For example, if $p = 3$, sample

$$\tilde{X}_1 \text{ from } \mathcal{L}(X_1 | X_2, X_3)$$

then sample

$$\tilde{X}_2 \text{ from } \mathcal{L}(X_2 | X_1, X_3, \tilde{X}_1)$$

and so on and so forth.

Crohn's Disease

Candès et al use the **knockoff procedure** to identify **gene clusters** that predict **Crohn's Disease** and compare the results to the **WTCCC study** published in *Nature* in 2007.

With $p = 71,145$ and $n = 14,708$, they're able to identify **18 gene clusters** that are predictive of Crohn's outcomes compared to **9 clusters** in the WTCCC paper.

Metropolis-Hastings

The Metropolis-Hastings algorithm (Metropolis et al 1953; Hastings 1970) provides a general **time-reversible Markov transition kernel** whose stationary distribution is an arbitrary density function π.

To construct a transition from x to y, generate x^* from a distribution $q(\cdot|x)$ and set

$$y = \begin{cases} x^* & \text{with prob } \alpha \\ x & \text{with prob } 1-\alpha \end{cases}$$

where

$$\alpha = \min\left(1, \frac{\pi(x^*)q(x|x^*)}{\pi(x)q(x^*|x)}\right)$$

The Target Distribution

As in *Panning for Gold*, the target distribution, i.e. π, is

$$\mathcal{L}\left(X_j \middle| X_{-j}, \tilde{X}_{1:j-1}\right).$$

However, the target density for this distribution cannot be evaluated because it depends on an integral over the proposal distribution, x^*. Hence, the target distribution, π, is set to be

$$\mathcal{L}\left(X_j \middle| X_{-j}, \tilde{X}_{1:j-1}, X^*_{1:j-1}\right).$$

Metropolized Knockoff Sampling

for $j = 1$ **to** p **do**

| Sample $X_j^* = x_j^*$ from a faithful proposal distribution q_j.

| Accept the proposal with probability

| $\min\left(1, \frac{q_j(x_j|x_j^*)\mathbb{P}\left(X_{-j}=x_{-j}, X_j=x_j^*, \tilde{X}_{1:j-1}=\tilde{x}_{1:j-1}, X^*_{1:j-1}=x^*_{1:j-1}\right)}{q_j(x_j^*|x_j)\mathbb{P}\left(X_{-j}=x_{-j}, X_j=x_j, \tilde{X}_{1:j-1}=\tilde{x}_{1:j-1}, X^*_{1:j-1}=x^*_{1:j-1}\right)}\right)$

| Upon acceptance, set $\tilde{x}_j = x_j^*$; otherwise, set $\tilde{x}_j = x_j$.

end

Knockoff Quality

Using *mean absolute correlation* as a measure of knockoff quality, where

$$\text{MAC}\left(\mathcal{L}\left(X, \tilde{X}\right)\right) := \frac{1}{p}\sum_{j=1}^{p}\left|\text{corr}\left(X_j, \tilde{X}_j\right)\right|$$

the Metropolized Knockoff Sampling algorithm takes **significantly longer to run** but results in knockoffs of **higher quality**.

Knockoffs from Copulas

Suppose F and F_i are the distribution functions of X and X_i respectively.

By *Sklar's Theorem*, there is a p-copula C such that, $\forall x \in \mathbb{R}^p$,

$$F(x) = C\left[F_1(x_1), \ldots, F_p(x_p)\right]$$

Then, for any symmetric 2-copulas $D_1, \ldots, D_p$, $\forall x \in \mathbb{R}^{2p}$,

$$H(x) = C\left[D_1\left(F_1(x_1), F_1(x_{p+1})\right), \ldots, D_p\left(F_p(x_p), F_p(x_{2p})\right)\right]$$

is a candidate for the distribution function of $(X, \tilde{X})$.

Is H a distribution function?

H is a distribution function on $\mathbb{R}^{2p}$ whenever

1. D_i is of class C^2 for each $i \in I$;
2. C has a density φ with respect to m_p, the Lebesgue measure on the Borel σ-field on $\mathbb{R}^p$;
3. φ is of class C^p and, at each point $u \in [0,1]^{2p}$, one obtains

$$\frac{\partial^p}{\partial u_{2p} \cdots \partial u_{p+1}} \varphi\big[D_1(u_1, u_{p+1}), \ldots, D_p(u_p, u_{2p})\big] \prod_{i=1}^{p} \frac{\partial}{\partial u_i} D_i(u_i, u_{p+i}) \geq 0$$

Copula Knockoffs General Algorithm

1. Choose $F_1, \ldots, F_p$ on $\mathbb{R}$, a p-copula C and a family of 2-copulas $D_1, \ldots, D_p$ in such a way that

$$H(x, \tilde{x}) = C\left[D_1\left(F_1(x_1), F_1(\tilde{x}_1)\right), \ldots, D_p\left(F_p(x_p), F_p(\tilde{x}_p)\right)\right]$$

is a distribution function on $\mathbb{R}^{2p}$.

2. Sample $\tilde{x}$ from the conditional density

$$p(\tilde{x}|x) = \frac{1}{\varphi[F_1(x_1), \ldots, F_p(x_p)] \prod_{i=1}^{p} f_i(x_i)} \times \frac{\partial^{2p} H}{\partial x_p \ldots \partial x_1 \partial \tilde{x}_p \ldots \partial \tilde{x}_1}(x, \tilde{x})$$

where φ and f_i are the densities of C and F_i respectively.

Copula Metropolis-Hastings Sampler

1. Choose the initial value $\tilde{x}^{(0)}$
2. Choose the proposal distribution $K(\cdot|x)$

for $j = 1$ **to** M **do**

3. Sample y from the proposal $K(\cdot|\tilde{x}^{(j-1)})$
4. Compute acceptance probability

$$\alpha(\tilde{x}^{(j-1)}, y) = \min\left\{ \frac{\frac{\partial^{2p} H}{\partial x_p \ldots \partial x_1 \partial \tilde{x}_p \ldots \partial \tilde{x}_1}(x, y)}{\frac{\partial^{2p} H}{\partial x_p \ldots \partial x_1 \partial \tilde{x}_p \ldots \partial \tilde{x}_1}(x, \tilde{x}^{(j-1)})} \frac{K(\tilde{x}^{(j-1)}|y)}{K(y|\tilde{x}^{(j-1)})} \right\}$$

5. Set $\tilde{x}^{(j)} = y$ with probability $\alpha(\tilde{x}^{(j-1)}, y)$ and $\tilde{x}^{(j)} = \tilde{x}^{(j-1)}$ with probability $1 - \alpha(\tilde{x}^{(j-1)}, y)$.

end

6. Return the sample $\tilde{x}^{(j)}, j = 1, \ldots, M$

The Challenge

We observe

$$\mathcal{X} \equiv (X_1, \ldots, X_p) \in \mathbb{R}^{p \times n}$$

where
p is the number of features and
n is the number of observations.

We want to create

$$\tilde{\mathcal{X}} \equiv (\tilde{X}_1, \ldots, \tilde{X}_p) \in \mathbb{R}^{p \times n}$$

such that $\tilde{\mathcal{X}}$ is a set of valid knockoffs for $\mathcal{X}$.

The Constraints

If

$$\mu_{m,k}(X_i, X_j)$$

is the mth co-moment between features i and j with balance parameter $k < m$, then for

$$2 \leq m \leq 4 \text{ and } 1 \leq k \leq (m-1)$$

our constraints are

- $\tilde{X}_i \overset{d}{=} X_i, \ \forall i$
- $\mu_{m,k}(\tilde{X}_i, X_j) = \mu_{m,k}(X_i, X_j), \ \forall i \neq j$
- $\mu_{m,k}(\tilde{X}_i, \tilde{X}_j) = \mu_{m,k}(X_i, X_j), \ \forall i \neq j$

The Optimisation

I tell the program to minimise the **summed squared correlation** between each feature and its knockoff:

$$\sum_{i=1}^{p} \left(\mu_{2,1}(\tilde{X}_i, X_i) \right)^2$$

subject to the constraints given.

To reduce the computation time as much as possible, I estimate the multivariate distribution using a Gaussian copula:

$$\Phi \left(\Phi^{-1} \left(F_1(X_1), \ldots, F_p(X_p) \right) \right)$$

and use the methodology set out in Candès et al (2018) to generate an initial guess.

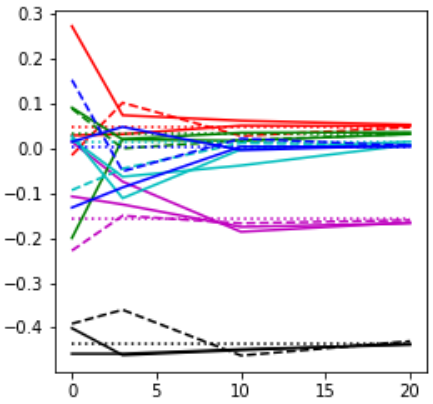
Artificial Features

To test the algorithm, I sample **four Normal(0,1)** random variables with **100 observations** each. I then shuffle them to artificially generate non-zero correlation, coskewness, and cokurtosis as follows

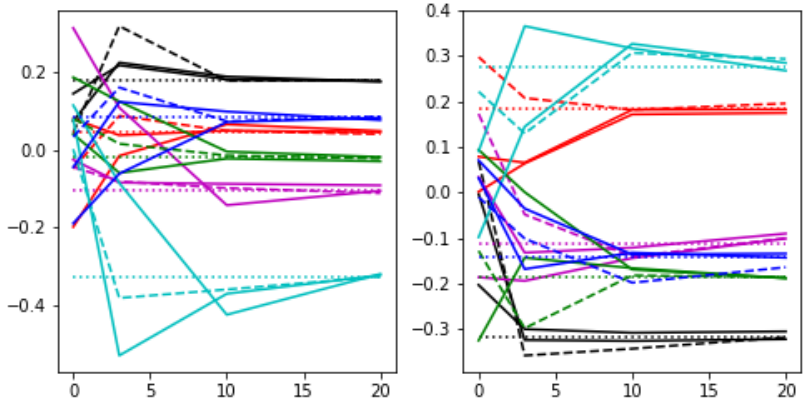
- Feature 1 has **smaller** observations shuffled towards the end of the sample,
- Feature 2 has **larger** observations shuffled towards the end of the sample,
- Feature 3 has observations **close to the mean** shuffled towards the end of the sample,
- Feature 4 is left **unshuffled**.

I run the algorithm for up to **20 iterations**.

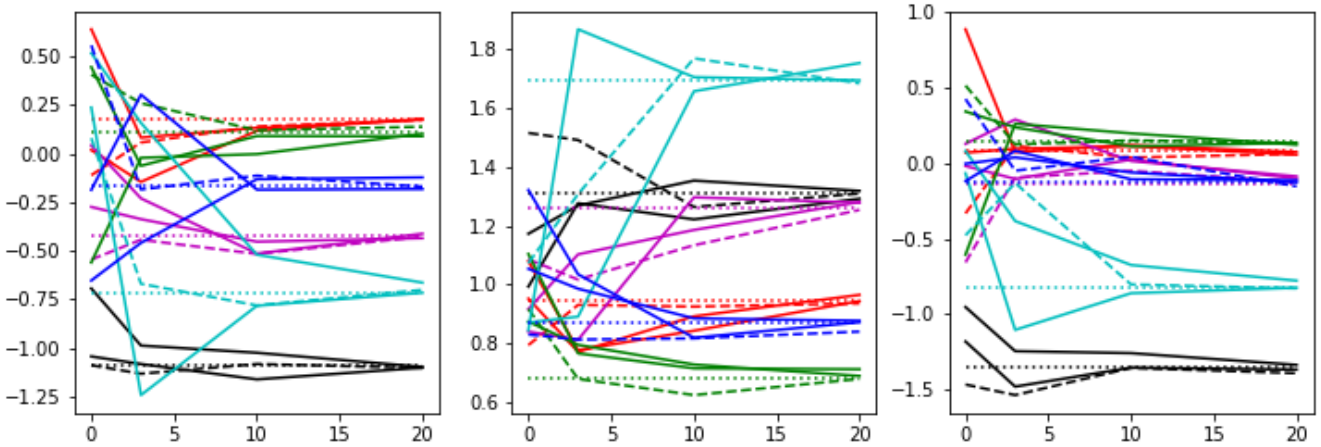
Model Performance



(a) Correlation

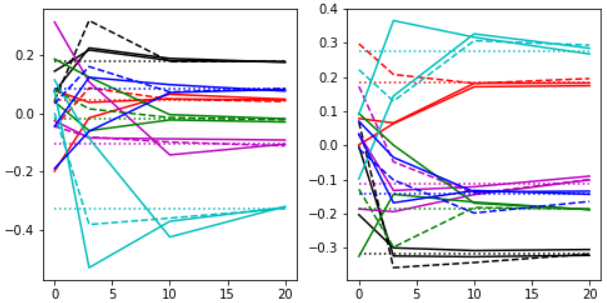


(b) Coskewness

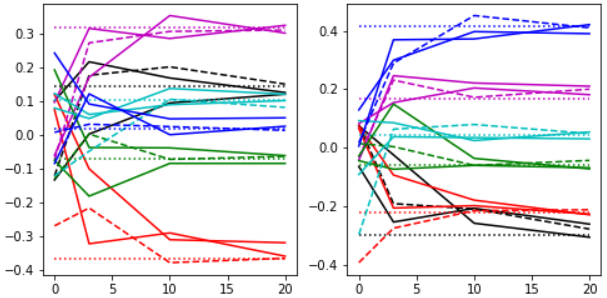


(c) Cokurtosis

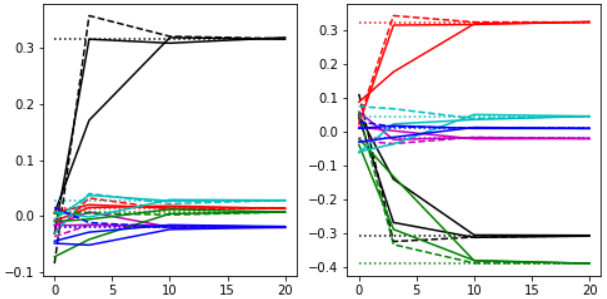
More Parameters



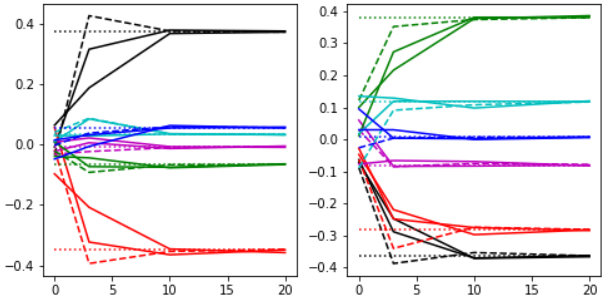
(a) $p = 4$, $n = 100$; -:–:26



(b) $p = 8$, $n = 100$; -:01:27



(c) $p = 4$, $n = 1000$; -:12:40



(d) $p = 8$, $n = 1000$; 1:26:15

Distributions and Correlations

Worst outcome	Dist. p-value	Corr. with Feature
$p = 4, n = 100$	35.8%	0.185
$p = 4, n = 1000$	12.9%	0.080
$p = 8, n = 100$	36.9%	0.298
$p = 8, n = 1000$	6.4%	0.099

Stock Data

I take daily returns data from the Swiss Market Index for the **20 constituents** for the period 26th May 2017 to the 8th April 2019. This sample contains **481 observations**.

I **scale** the returns so that they all have **mean 0** and **variance 1** and then estimate the joint distribution using a Gaussian copula.

This process identifies that the univariate distributions of all the stocks are **student-t** with degrees of freedom ranging from 3.08 to 5.81.

Results

I run the algorithm for **100 iterations** for a total time of **13:37:27**.

Quartiles of the **20** target variables:

Quartile	Min Lonza	0.25	0.50	0.75	Max UBS
Corr.	0.36	0.51	0.60	0.71	0.92

Contrainst Adherence

		Nestlé			UBS		
		Real	Feature	Knockoff	Real	Feature	Knockoff
Lonza	Correlation	0.364	0.366	0.365	0.424	0.424	0.424
	Left Coskewness	-0.030	-0.026	-0.030	-0.122	-0.122	-0.120
	Right Coskewness	-0.155	-0.157	-0.157	-0.273	-0.274	-0.275
	Left Cokurtosis	1.181	1.178	1.179	1.642	1.643	1.643
	Central Cokurtosis	1.579	1.577	1.574	1.871	1.872	1.875
	Right Cokurtosis	1.417	1.432	1.420	1.811	1.805	1.807
Nestlé	Correlation				0.337	0.336	0.336
	Left Coskewness				-0.136	-0.134	-0.136
	Right Coskewness				-0.233	-0.237	-0.236
	Left Cokurtosis				1.599	1.596	1.605
	Central Cokurtosis				1.793	1.796	1.786
	Right Cokurtosis				1.437	1.434	1.444

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The End

Questions? Comments?